

Automorphism invariance for arbitrary increasing radial weights

A disproof of Conjecture 2 in arXiv:2502.05533

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Status. Candidate counterexample/disproof; proof judged likely valid and awaiting human review.

1 Source question

Norrbo's paper [1] defines, for a positive radial weight v and $1 \leq p < \infty$,

$$\gamma_p(f, a) = \|f \circ \sigma_a - f(a)\|_{H^p}, \quad \|f\|_{*,v,p} = \sup_{a \in \mathbb{D}} v(a) \gamma_p(f, a),$$

where $\sigma_a(z) = (a - z)/(1 - \bar{a}z)$. The space $\text{BMOA}_{v,p}$ consists of the $f \in H^p$ for which this seminorm is finite; BMOA_v denotes the case $p = 2$.

Conjecture 2 on page 39 predicts that some increasing radial weight fails to preserve membership under composition by a disk automorphism:

There exists an increasing, radial (not admissible) weight $v : \mathbb{D} \rightarrow (0, \infty)$, $\hat{\phi} \in \text{Aut}(\mathbb{D})$ and $f \in \text{BMOA}_v$ such that $f \circ \hat{\phi} \notin \text{BMOA}_v$.

Conjecture 2. There exists an increasing, radial (not admissible) weight $v : \mathbb{D} \rightarrow]0, \infty[$, $\hat{\phi} \in \text{Aut}$ and $f \in \text{BMOA}_v$ such that $f \circ \hat{\phi} \notin \text{BMOA}_v$.

A direct approach yields that this is the same as proving that there exists $\hat{\phi} \in \text{Aut}$ and $f \in \text{BMOA}_v$ such that

$$\sup_{a \in \mathbb{D}} \frac{v(\hat{\phi}(a))}{v(a)} v(a) \gamma(f, a, 2) = \infty.$$

Examining the quotient $\frac{v(\hat{\phi}(a))}{v(a)}$, we have the following: Using $v(z) = g(\frac{1}{1-|z|^2})$, for some increasing unbounded function $g : [1, \infty[\rightarrow]0, \infty[$, the automorphisms $\hat{\phi}_c := \sigma_{-\sqrt{1-c}}$, $0 < c < 1$ yield

$$(7.1) \quad g\left(\frac{1}{1 - |\hat{\phi}_c(a)|^2}\right) = g\left(\frac{|1 + \sqrt{1-c}|^2}{c(1 - |a|^2)}\right) \geq g\left(\frac{1}{c} \frac{1}{1 - |a|^2}\right), \quad a \in]0, 1[.$$

For any increasing unbounded function $G : [1, \infty[\rightarrow [1, \infty[$ of any growth rate, we can create a continuous g , dominated by G , as follows: Let $x_0 = 1$ and $g(x) = 1$ for all $x \in [x_0, x_1]$, where $x_1 > x_0$ is a point where $G(x_1) \geq 1$. On $[x_1, x_1 + 1]$ we define g to increase to the value $G(x_1 + 1)$. Define g to be constant for all $x \in [x_1 + 1, x_2]$, where $x_2 > x_1 + 1$ is such that $G(x_2)/G(x_1 + 1) \geq 2$. Continuing this process we obtain a sequence (x_n) , with the property

Figure 1: Conjecture 2 and its equivalent divergence formulation in the source paper [1, p. 39].

The theorem below proves that no such example exists.

2 Proof intuition

The oscillation $\gamma_p(f, a)$ is stable when its center moves a bounded pseudohyperbolic distance. A fixed disk automorphism changes $1 - |a|$ only by a constant factor, although it may rotate the center by nearly π . If $b = \phi(a)$ lies farther inside the disk than a , replace b by the point c on the same ray with $|c| = |a|$. The comparable boundary depths make b and c pseudohyperbolically close. Thus $\gamma_p(f, b)$ is controlled by $\gamma_p(f, c)$, and radiality gives $v(c) = v(a)$. If b lies farther out than a , monotonicity gives $v(a) \leq v(b)$ directly. These two cases exhaust the disk.

3 Formal result

Write

$$\|f\|_{\text{BMOA}_{v,p}} = |f(0)| + \|f\|_{*,v,p}.$$

Lemma 1 (local comparison of oscillations). *For $1 \leq p < \infty$ and $0 \leq q < 1$, there is a constant $M_{p,q}$ such that*

$$\gamma_p(f, z) \leq M_{p,q} \gamma_p(f, w)$$

for every $f \in H^p$ and every $z, w \in \mathbb{D}$ satisfying

$$\rho(z, w) := |\sigma_w(z)| \leq q.$$

Proof. Put $\eta = \sigma_w(z)$ and

$$g = f \circ \sigma_w - f(w).$$

Then $g(0) = 0$, $\|g\|_{H^p} = \gamma_p(f, w)$, and automorphism covariance of γ_p gives

$$\gamma_p(f, z) = \gamma_p(g, \eta).$$

For $|\eta| \leq q$, the boundary change-of-variables formula for σ_η and the estimate

$$P_\eta(\zeta) \leq \frac{1+q}{1-q} \quad (\zeta \in \mathbb{T})$$

give

$$\|g \circ \sigma_\eta\|_{H^p} \leq \left(\frac{1+q}{1-q} \right)^{1/p} \|g\|_{H^p}.$$

The Hardy point-evaluation estimate also gives

$$|g(\eta)| \leq (1 - q^2)^{-1/p} \|g\|_{H^p}.$$

Minkowski's inequality now yields

$$\gamma_p(g, \eta) \leq \left[\left(\frac{1+q}{1-q} \right)^{1/p} + (1 - q^2)^{-1/p} \right] \|g\|_{H^p}.$$

This proves the lemma. □

Theorem 1 (unconditional automorphism invariance). *Let $1 \leq p < \infty$, and let $v : \mathbb{D} \rightarrow (0, \infty)$ be radial and nondecreasing in the radius. For every $\phi \in \text{Aut}(\mathbb{D})$, composition induces bounded isomorphisms*

$$C_\phi : \text{BMOA}_{v,p} \longrightarrow \text{BMOA}_{v,p} \quad \text{and} \quad C_\phi : \text{VMOA}_{v,p} \longrightarrow \text{VMOA}_{v,p}.$$

No admissibility, doubling, or regular-growth assumption on v is required.

Proof. Write

$$\phi(z) = \lambda \frac{\alpha - z}{1 - \bar{\alpha}z}, \quad |\lambda| = 1, \quad \alpha \in \mathbb{D},$$

and set

$$K = \frac{1 + |\alpha|}{1 - |\alpha|}.$$

The standard automorphism identity gives, for $b = \phi(a)$,

$$\frac{1 - |b|^2}{1 - |a|^2} = \frac{1 - |\alpha|^2}{|1 - \bar{\alpha}\alpha|^2} \in [K^{-1}, K].$$

Since $1 \leq 1 + |z| \leq 2$, it follows that

$$\frac{1 - |b|}{1 - |a|} \in [(2K)^{-1}, 2K] \quad (a \in \mathbb{D}). \quad (1)$$

We also use exact automorphism covariance:

$$\gamma_p(f \circ \phi, a) = \gamma_p(f, \phi(a)). \quad (2)$$

Indeed, $\phi \circ \sigma_a$ and $\sigma_{\phi(a)}$ differ only by precomposition with a rotation, and the H^p norm is rotation invariant.

Fix $a \in \mathbb{D}$ and put $b = \phi(a)$. There are two cases.

If $|a| \leq |b|$, monotonicity gives

$$v(a)\gamma_p(f, b) \leq v(b)\gamma_p(f, b) \leq \|f\|_{*,v,p}.$$

Suppose $|a| > |b|$. Let c be the point on the ray from 0 through b with $|c| = |a|$; if $b = 0$, choose any argument for c . Put $s = |b|$ and $t = |c| = |a|$. From (1),

$$\frac{1 - t}{1 - s} \geq m := \frac{1}{2K}.$$

Because b and c lie on the same radius,

$$\rho(b, c) = \frac{t - s}{1 - st} = \frac{(1 - s) - (1 - t)}{(1 - s) + (1 - t) - (1 - s)(1 - t)} \leq 1 - \frac{1 - t}{1 - s} \leq 1 - m < 1.$$

The same bound holds when $b = 0$. Lemma 1, radiality, and $|c| = |a|$ therefore give

$$v(a)\gamma_p(f, b) = v(c)\gamma_p(f, b) \leq M_{p,1-m} v(c)\gamma_p(f, c) \leq M_{p,1-m} \|f\|_{*,v,p}.$$

Together with (2), the two cases show

$$\|f \circ \phi\|_{*,v,p} \leq C_{p,\phi} \|f\|_{*,v,p}.$$

The full norm is also controlled. At $a = 0$, the seminorm controls the nonconstant H^p part, so

$$\|f\|_{H^p} \leq |f(0)| + v(0)^{-1} \|f\|_{*,v,p}.$$

Composition by the fixed automorphism ϕ and evaluation at the fixed point $\phi(0)$ are bounded on H^p . Hence C_ϕ is bounded on $\text{BMOA}_{v,p}$. Applying the same argument to ϕ^{-1} proves that it is an isomorphism.

Finally, suppose $f \in \text{VMOA}_{v,p}$ and $|a| \rightarrow 1$. Formula (1) implies $|b| = |\phi(a)| \rightarrow 1$. In the first case above, the controlling quantity is $v(b)\gamma_p(f, b) \rightarrow 0$. In the second case, $|c| = |a| \rightarrow 1$, and the controlling quantity is $v(c)\gamma_p(f, c) \rightarrow 0$. Thus $f \circ \phi \in \text{VMOA}_{v,p}$; applying the result to ϕ^{-1} again gives an isomorphism. $\square$

4 Consequence for the source conjecture

Taking $p = 2$ in Theorem 1 shows that for every increasing radial weight v , every $\widehat{\phi} \in \text{Aut}(\mathbb{D})$, and every $f \in \text{BMOA}_v$, one has

$$f \circ \widehat{\phi} \in \text{BMOA}_v.$$

This is the logical negation of the existence statement in Conjecture 2. Therefore Conjecture 2 is false.

5 Verification and limitations

The proof is self-contained and uses no computational or conditional step. The local oscillation lemma, the boundary-depth comparison, the case $\phi(a) = 0$, the inverse automorphism, and the vanishing condition were checked separately. The result concerns only Conjecture 2; the other conjectures in the source paper are untouched.

A bounded novelty search on 27 August 2026 covered the run’s lightweight indexes, the exact arXiv identifier and paper title, and web queries combining “weighted BMOA”, “increasing radial weight”, and “automorphism composition”. It found the source preprint and 2025 journal article but no later solution or disproof of Conjecture 2. This was not an exhaustive publication-level search.

6 Human-review recommendation

Review as a full disproof of Conjecture 2. The most important checks are equations (2) and (1), followed by the radial comparison point in the second case of the proof.

References

- [1] D. Norrbo, *Compactness and related properties of weighted composition operators on weighted BMOA spaces*, Bulletin des Sciences Mathématiques **203** (2025), 103642; arXiv:2502.05533.